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SUB-SOIL INVESTIGATION REPORT

FOR THE PROPOSED
FIVE STORIED RESIDENTIAL BUILDING

At Shahporan, Sadar, Sylhet.
Dag # Sa-167, Bs-823, Kha # Bs-1596, Mutation-2020
J.L- Sa-66, Bs-55, Mouja- Chuabohor Boteshwar.

CLIENT

ABUL KALAM AZAD & GONG

Verification Code: 4MSJsL1c
Verify report at: crtc-cee-sust.com

CONDUCTED BY

CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET

SAMPLE COLLECTED BY

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Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would necessary to select proper type of foundation.

Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

Table – 1: Items of the supplied samples

No	Name	Bore hole no	Depth	Quantities
01	Disturbed sample	01	0-35 ft	07
02	Disturbed sample	02	0-45 ft	09
03	Disturbed sample	03	0-45 ft	09
04	Disturbed sample	04	0-50 ft	10
05	Disturbed sample	05	0-45 ft	09
06	-	-	-	-
07	-	-	-	-
08	-	-	-	-
09	-	-	-	-
10	-	-	-	-

Laboratory Classifications

The disturbed soil samples, supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with bore hole (BH) and depth (Please see enclosed Tables).

Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)

N-values	Consistency	Unconfined Compressive Strength (q_u)	
		tsf	kPa
0-2	Very soft	0 to 0.25	0 to 25
3-5	Soft	0.25 to 0.50	25 to 50
6-9	Medium	0.50 to 1	50 to 100
10-16	Stiff	1 to 2	100 to 200
17-30	Very stiff	2 to 4	200 to 400
>30	Hard	>4	>400

Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
1 – 2	Very Loose	0 – 15	26 – 28	70 – 100
3 – 6	Loose	15 – 35	28 – 30	90 – 115
7 – 15	Medium	35 – 65	30 – 34	110 – 130
16 – 30	Dense	65 – 85	33 – 38	110 – 140
>30	Very Dense	>85	>50	130 – 150

Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
2 – 3	Very Loose	0 – 15	27 – 28	70 – 100
4 – 7	Loose	15 – 35	30 – 32	90 – 115
8 – 20	Medium	35 – 65	32 – 36	110 – 130
21 – 40	Dense	65 – 85	36 – 42	110 – 140

Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
3 – 6	Very Loose	0 – 15	28 – 30	70 – 100
5 – 9	Loose	15 – 35	30 – 34	90 – 115
10 – 25	Medium	35 – 65	33 – 40	110 – 130
26 – 40	Dense	65 – 85	40 – 50	110 – 140
>40	Very Dense	>85	>50	130 – 150

<u>Bearing capacity for deep foundation</u>

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

Table 6: Formula Used to Determine Bearing Capacity of Soil

Shallow Foundation	
For Cohesive Soil	
$q_a = q_u + \gamma \frac{D}{3}$ $q_a = q_u$(Common Practice)	Where, q_a=Allowable bearing capacity, q_u= Unconfined Compressive Strength, γ= Unit Weight of soil, D= Depth of footing
For Cohesion less Soil	
$q_{ult} = 30N$; $q_a = q_{ult}/FS$	q_{ult}= Ultimate bearing capacity in kPa, N = SPT Number, FS= Factor of Safety
Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition	
Deep Foundation	
For Cohesive Soil	
$f_{su} = \alpha C_u$; $q_{pu} = N_c C_u$	α = Adhesion factor, f_{su} = Ultimate skin friction, C_u = Undrained Shear Strength, q_{pu} = Ultimate end bearing of pile, N_c = Bearing capacity factor for deep foundation=9
For Cohesion less Soil	
$f_{su} = N/50\ tsf$; $q_{pu} = 4N\ tsf$	f_{su} = Ultimate skin friction, q_{pu} = Ultimate end bearing of pile, N = SPT Number
Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"	

Based on the above formula, the important parameters for foundation design have been determined and are given in **Table 7**, **Table 8, Table 9, Table 10 and Table 11** respectively.

Table 7: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5)

Bore Hole No.	**Depth**	**Design N**	**Clay**						**Sand**
			Continuous			**Square(Isolated)**			
	(ft)	**-**	**D/B=0.25**	**D/B=0.5**	**D/B=1**	**D/B=0.25**	**D/B=0.5**	**D/B=1**	
BH01	**2.5-5.0**	**10**	1.206	1.269	1.383	1.425	1.539	1.674	1.300
	5.0-10	**11**	1.328	1.397	1.523	1.568	1.694	1.843	1.431
BH02	**2.5-5.0**	**3**	0.371	0.390	0.426	0.438	0.474	0.515	0.400
	5.0-10	**7**	0.792	0.833	0.908	0.935	1.010	1.099	0.853
BH03	**2.5-5.0**	**3**	0.348	0.366	0.399	0.411	0.444	0.483	0.375
	5.0-10	**6**	0.712	0.749	0.816	0.841	0.908	0.988	0.767
BH04	**2.5-5.0**	**3**	0.394	0.415	0.452	0.466	0.503	0.547	0.425
	5.0-10	**6**	0.708	0.745	0.812	0.836	0.903	0.983	0.763
BH05	**2.5-5.0**	**3**	0.348	0.366	0.399	0.411	0.444	0.483	0.375
	5.0-10	**6**	0.672	0.707	0.770	0.794	0.857	0.933	0.724
BH06	**2.5-5.0**	**0**	-	-	-	-	-	-	-
	5.0-10	**0**	-	-	-	-	-	-	-
BH07	**2.5-5.0**	**0**	-	-	-	-	-	-	-
	5.0-10	**0**	-	-	-	-	-	-	-
BH08	**2.5-5.0**	**0**	-	-	-	-	-	-	-
	5.0-10	**0**	-	-	-	-	-	-	-
BH09	**2.5-5.0**	**0**	-	-	-	-	-	-	-
	5.0-10	**0**	-	-	-	-	-	-	-
BH10	**2.5-5.0**	**0**	-	-	-	-	-	-	-
	5.0-10	**0**	-	-	-	-	-	-	-

Table 8: The allowable skin friction for Precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.268	0.138	0.118	0.118	0.156	-	-	-	-	-
10	0.174	0.049	0.049	0.074	0.049	-	-	-	-	-
15	0.174	0.074	0.074	0.074	0.049	-	-	-	-	-
20	0.144	0.120	0.104	0.096	0.104	-	-	-	-	-
25	0.168	0.136	0.120	0.096	0.104	-	-	-	-	-
30	0.232	0.144	0.112	0.112	0.120	-	-	-	-	-
35	0.224	0.176	0.136	0.136	0.160	-	-	-	-	-
40	-	0.669	0.216	0.216	0.216	-	-	-	-	-
45	-	0.669	0.208	0.208	0.208	-	-	-	-	-
50	-	-	-	0.200	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 9: The allowable end bearing for Precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	4.500	1.350	1.125	1.125	1.575	-	-	-	-	-
10	1.800	0.450	0.450	0.675	0.450	-	-	-	-	-
15	1.800	0.675	0.675	0.675	0.450	-	-	-	-	-
20	28.800	24.000	20.800	19.200	20.800	-	-	-	-	-
25	33.600	27.200	24.000	19.200	20.800	-	-	-	-	-
30	46.400	28.800	22.400	22.400	24.000	-	-	-	-	-
35	44.800	35.200	27.200	27.200	32.000	-	-	-	-	-
40	-	11.250	43.200	43.200	43.200	-	-	-	-	-
45	-	11.250	41.600	41.600	41.600	-	-	-	-	-
50	-	-	-	40.000	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 10: The allowable skin friction for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.174	0.090	0.077	0.077	0.101	-	-	-	-	-
10	0.113	0.032	0.032	0.048	0.032	-	-	-	-	-
15	0.113	0.048	0.048	0.048	0.032	-	-	-	-	-
20	0.094	0.078	0.068	0.062	0.068	-	-	-	-	-
25	0.109	0.088	0.078	0.062	0.068	-	-	-	-	-
30	0.151	0.094	0.073	0.073	0.078	-	-	-	-	-
35	0.146	0.114	0.088	0.088	0.104	-	-	-	-	-
40	-	0.435	0.140	0.140	0.140	-	-	-	-	-
45	-	0.435	0.135	0.135	0.135	-	-	-	-	-
50	-	-	-	0.130	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 11: The allowable end bearing for Bored piles in Tsf (FS=3.0)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	2.025	0.608	0.506	0.506	0.709	-	-	-	-	-
10	0.810	0.203	0.203	0.304	0.203	-	-	-	-	-
15	0.810	0.304	0.304	0.304	0.203	-	-	-	-	-
20	12.960	10.800	9.360	8.640	9.360	-	-	-	-	-
25	15.120	12.240	10.800	8.640	9.360	-	-	-	-	-
30	20.880	12.960	10.080	10.080	10.800	-	-	-	-	-
35	20.160	15.840	12.240	12.240	14.400	-	-	-	-	-
40	-	5.063	19.440	19.440	19.440	-	-	-	-	-
45	-	5.063	18.720	18.720	18.720	-	-	-	-	-
50	-	-	-	18.000	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

CONCLUSION AND RECOMMENDATION

On the basis of analysis and information provided by the owner following recommendations are suggested for the proposed five storied residential building.

Soil profile scenario

Overall soil information of the investigated site is quite irregular in between the boreholes and vertical inconsistency exists in the soil formation. For boreholes 1 and 2, brown stiff clay with good bearing capacity exists roughly up to top 5 ft from EGL followed by a brown soft clay layer with low bearing capacity. After that, from 20 ft depth, grey fine sand layer have been observed up to 35 ft (borehole 1 and 2) and then a grey hard clay layer have been found (borehole 2). For boreholes 3 and 4 brown sandy clay with low bearing capacity exists roughly up to top 5 ft from EGL followed by a brown soft clay layer. From 20 ft depth, a grey fine sand layer have been observed up to the whole depth of the boreholes. For borehole 5, grey sandy clay with low bearing capacity exists roughly up to top 5 ft from EGL followed by a brown soft clay layer. From 20 ft depth, a grey fine sand layer have been observed up to the whole depth of the borehole as client provided. From the above discussion it is very clear that the whole scenario of the proposed site is quite variable in nature.

Foundation options

In this situation, the foundation should be selected considering both the dimension and design load for the proposed building. Since soft clay soil with low bearing capacity is found in the top 15 ft layer, proper ground improvement is needed before adopting any shallow foundation. In this regard, the soft soil from the top 10 ft depth (only in the footing area) is to be excavated first and then a compacted bed of minimum 5 ft thickness should be prepared using brick/stone chips and sand. Also, geotextile could be used at the bottom of the compacted bed. The bearing capacity for shallow foundation which is given in **table 7** is based on existing soil conditions. However, client needs to reassess the improved bearing capacity of the prepared ground after implementing soil improvement technique in the top layer as mentioned above.

Safety provisions

It is strongly recommended to check the **settlement** of the site considering building load while designing either shallow or deep foundation. Obviously **pile load test** covering the entire project area must be carried out for confirming the design parameters of the service pile (*if pile foundation is considered*). Moreover, the earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be taken into account for the safe design of shallow or deep foundation.

Attachments

The layout of the bore holes of the investigated site and the bore-log profiles with SPT values is shown in the Appendix. Client performed no special laboratory test for the proposed site.

Special notes

These suggestions are absolutely tentative and selection of type, depth and dimensions of foundation would be decided by the Consultant or Design Engineer. It is to be noted that the recommendation is restricted to the existing soil condition of **35-50** ft as client provided.

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